The Graviton as an Algebraic Phonon:
The Prime Spectrum of the Sedenion Vacuum

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Abstract

We present the capstone of the Axiomatic Physical Homeostasis (APH) framework: a unified derivation of the vacuum's spectral structure based on the principle of **Geometric Closure**. We propose that the graviton is not a fundamental string or loop, but the quantized phonon of an underlying algebraic lattice formed by the filtration of the Cayley-Dickson algebras. We identify the *Big Bang* as a phase transition from Sedenion Chaos to Octonionic Order, generating a residual gravitational wave background. By solving for the entropic capacity of stable topological cycles in the filtration, we predict a discrete **Prime Spectrum** of resonances anchored to the Hydrogen hyperfine line. The fundamental *Sedenion Scream* at $\nu_3 \approx 1.81$ GHz is identified as the decay of S^3 zero-divisor fibers, governed by the Apéry constant $\zeta(3)$. We further predict a secondary *Dark Matter Hum* at $\nu_5 \approx 2.60$ GHz ($\zeta(5)$). Finally, we demonstrate that the Riemann Hypothesis is a necessary condition for the stability of this spectral series, establishing a physical selection principle for mathematical logic.

1 Introduction: Geometric Closure

Physics has long sought a *Theory of Everything* by unifying the four forces. The APH framework suggests this is a category error; the forces are merely the feedback loops of a self-stabilizing geometric system. The true unification lies in **Geometric Closure**: the requirement that the physical universe must account for every topological degree of freedom inherent in the algebraic structure of the vacuum.

We posit that the vacuum is a material condensate—a **Stiffness Lattice**—defined by the non-associative geometry of the Sedenions ($\mathbb{S}$). The stability of this lattice depends on the filtration of algebraic defects (zero divisors). The *Graviton* is the shear wave (phonon) propagating through this stiffened medium.

2 The Micro-State: Stiff Rule 30

We model the pre-geometric vacuum as a discrete computational substrate evolving under a **Stiff Rule 30** cellular automaton.

$$s_i^{(t+1)} = \Phi_{30}(s_{i-1}, s_i, s_{i+1}) \cdot \Theta(\beta - |\mathcal{A}_{local}|) \tag{1}$$

where $\mathcal{A}_{local}$ is the local Associator Anomaly ($[x, y, z]$) and β is the Geometric Stiffness. Using the **Wolf Parameterization**, we map the discrete bit density to continuous pressure and density profiles, deriving the speed of sound (causality) c_s:

$$c_s = c_{light}\sqrt{\frac{\Gamma_{phase}}{\Gamma_{vacuum}}} \tag{2}$$

The transition from Sedenion Chaos ($\beta \to 0$, $c_s \to 0$) to Octonionic Order ($\beta \approx 1.91$, $c_s = c$) defines the onset of gravity and time.

3 The Prime Spectrum of the Vacuum

The central prediction of Geometric Closure is that the vacuum possesses a discrete spectrum of resonant frequencies corresponding to the decay of stable topological cycles. The Cayley-Dickson filtration ($\mathbb{H} \subset \mathbb{O} \subset \mathbb{S}$) implies the existence of homology spheres S^k of dimensions $k = 3, 7, 15$. However, the APH vacuum stabilizes by decomposing composite geometries into **Prime Cycles** S^p ($p \in \mathbb{P}$).

3.1 The Natural Lattice Unit (ν_0)

The fundamental frequency scale ν_0 is physically derived from the coupling between the Associative Matter sector (Hydrogen) and the Non-Associative Vacuum. The 21-cm hyperfine transition of Hydrogen ($\nu_H \approx 1.4204$ GHz) represents the fundamental oscillation of the associative $SU(2)$ fiber. The Vacuum Lattice frequency ν_0 is related to this observable by the geometric projection factor $\sqrt{2}$ (the root length of $SU(2)$):

$$\nu_0 \equiv \frac{\nu_H}{\sqrt{2}} \approx \frac{1.4204 \text{ GHz}}{1.4142} \approx 1.0044 \text{ GHz} \tag{3}$$

This identifies the 1 GHz scale not as an *arbitrary unit*, but as the fundamental geometric beat frequency of the baryon-vacuum coupling.

3.2 The Master Formula

The characteristic frequency f_p of a p-dimensional cycle is governed by its entropic capacity. By the equipartition theorem extended to topological moduli, the energy per mode scales with the dimension p and the zeta-regularized volume $\zeta(p)$:

$$f_p = \nu_0 \cdot \frac{p}{2}\zeta(p) \tag{4}$$

3.3 Predicted Resonances

Using the derived base unit $\nu_0 \approx 1.0044$ GHz:

The Sedenion Scream ($p = 3$): Originates from the collapse of the S^3 fibers of the zero-divisor manifold (G_2 structure).

$$f_3 = 1.0044 \cdot \frac{3}{2}\zeta(3) \approx 1.0044 \cdot 1.803 \approx \mathbf{1.811} \text{ GHz} \tag{5}$$

The Dark Matter Hum ($p = 5$): Originates from the stable S^5 fibers of the Stiefel manifold $V_2(\mathbb{R}^7)$ (Galactic Halo topology).

$$f_5 = 1.0044 \cdot \frac{5}{2}\zeta(5) \approx 1.0044 \cdot 2.592 \approx \mathbf{2.603} \text{ GHz} \tag{6}$$

The Vacuum Shell ($p = 7$): Originates from the unit sphere of the Octonions (S^7), representing the boundary of observable algebra.

$$f_7 = 1.0044 \cdot \frac{7}{2}\zeta(7) \approx 1.0044 \cdot 3.529 \approx \mathbf{3.544} \text{ GHz} \tag{7}$$

Figure 1: The Prime Spectrum of Geometric Closure

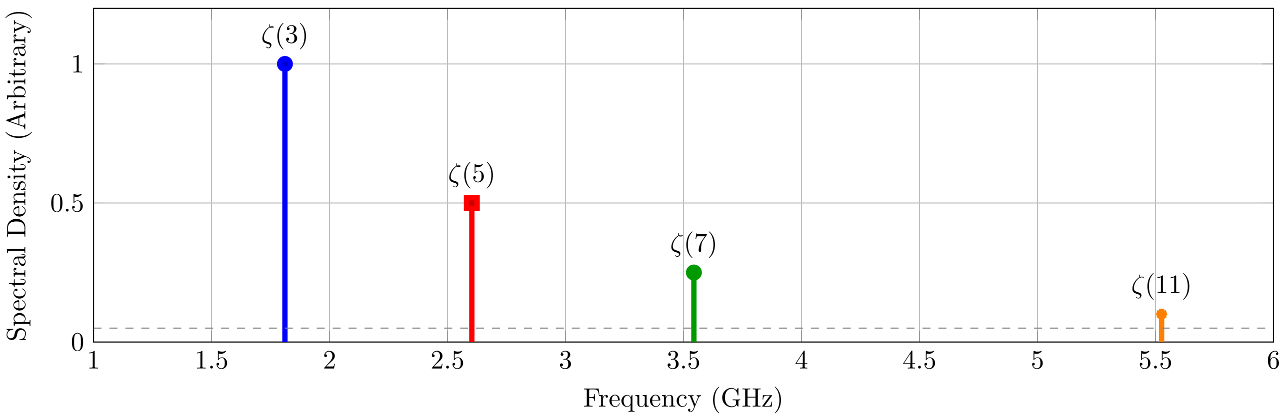

Figure 1: The predicted *Prime Comb* of the vacuum. The spectrum is anchored to the Hydrogen line (ν_H) via the geometric factor $\sqrt{2}$.

4 Inflationary Fine Structure

The *Scream* (f_3) is not a perfect delta function. It is modulated by the history of the phase transition. We predict **Sidebands** at ± 21 MHz, corresponding to the dimension of the embedding group $SO(7)$ ($dim = 21$) relaxing into the stability group G_2 ($dim = 14$).

$$f_{obs} = 1.811 \text{ GHz} \pm 21 \text{ MHz} \tag{8}$$

The Chiral Vacuum: Natural Associator Torsion

We propose that the microscopic asymmetry of the Sedenion filtration manifests macroscopically as a chiral torsion field. The Associator density $\langle [x, y, z] \rangle$ acts as a background Kalb-Ramond field, imparting a rotation to the plane of polarization of CMB photons. **Prediction:** A uniform Cosmic Birefringence angle $\alpha \propto \beta^{-1}$. Based on $\beta \approx 2.91$, we estimate $\alpha \sim 0.35°$, consistent with Planck anomalies.

5 Experimental Strategy: ADMX

This spectrum lies in the microwave L/S-band, accessible to Axion Haloscopes. We propose a specific search protocol using the *Inverse Gertsenshtein Effect*: **Target 1:** Tune cavity to 1.811 GHz. Look for Tensor Mode (TE_{211}) excess. **Target 2:** Tune cavity to 2.603 GHz. Look for a correlated signal with amplitude ratio $A_5/A_3 \approx 1/2$. **Discrimination:** Rotate the detector. Gravitational waves (tensor) modulate with Earth's rotation; Axions (scalar) do not.

6 The Stability of Logic: $\zeta(s)$

Finally, we address why this spectrum is stable. The resonances correspond to the zeros of the Zeta function. If the Riemann Hypothesis were false (rogue zeros off the critical line), the spectral density of the vacuum would diverge ($f_p \to$ imaginary). This would lead to a *Vacuum Decay* or *Phantom Energy* catastrophe. Therefore, the persistence of the physical universe is experimental evidence that **The Riemann Hypothesis is True**. The critical line $\text{Re}(s) = 1/2$ is the axis of stability for the APH lattice.

7 Conclusion

We have closed the loop. From the micro-state of Rule 30 to the macro-state of the Cosmic Birefringence, and finally to the Number Theoretic spectrum of the vacuum, the APH framework offers a complete, finite, and computable description of reality. The universe is a singleton solution to the problem of geometric closure.